

Ex 6 Time Series Analysis of Website Traffic

```
from google.colab import files
from PIL import Image
import matplotlib.pyplot as plt

uploaded = files.upload()

filename = list(uploaded.keys())[0]

original_img = Image.open(filename)

def apply_augmentations(img):
    rotated_img = img.rotate(45, expand=True)
    transposed_img = img.transpose(Image.FLIP_LEFT_RIGHT)
    resized_img = img.resize((200, 200))

    return rotated_img, transposed_img, resized_img

img_rot, img_trans, img_res = apply_augmentations(original_img)

fig, axes = plt.subplots(1, 4, figsize=(16, 4))

axes[0].imshow(original_img)
axes[0].set_title(f"Original\n{original_img.size}")
axes[0].axis("off")

axes[1].imshow(img_rot)
axes[1].set_title(f"Rotated (45°)\n{img_rot.size}")
axes[1].axis("off")

axes[2].imshow(img_trans)
axes[2].set_title(f"Flipped\n{img_trans.size}")
axes[2].axis("off")

axes[3].imshow(img_res)
axes[3].set_title(f"Resized\n{img_res.size}")
axes[3].axis("off")

plt.tight_layout()
plt.show()
```

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Saving nature.jpg to nature (2).jpg

Original
(447, 447)



Rotated (45°)
(633, 633)



Flipped
(447, 447)



Resized
(200, 200)

